Artificial Intelligence for Cybersecurity and Intrusion Detection

Lecture Notes and Solved Exercises

Dr Sarra ARAAR



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About the Book

Artificial intelligence can support the detection of suspicious activity in increasingly complex digital networks. This book introduces machine learning and deep learning techniques for cybersecurity, from data preprocessing and model evaluation to the design of intrusion detection systems. It examines model selection, explainable AI, and federated learning, then applies these concepts to a case study on smart-grid security. A final chapter surveys Transformers, graph neural networks, foundation models, and agentic AI. Organised as lecture notes, the chapters pair theoretical explanations with worked examples, graded exercises, and detailed solutions, enabling readers to examine both methodological choices and operational constraints.

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